Credit Risk Portfolio Analytics — Solution Overview

A governed monitoring workflow for portfolio analysts

Dashboard workflow

PORTFOLIO
KPIs, exposure, risk bands, cohort trends and income/employment/region/age/loan cuts. Filters drive every chart.

ACCOUNT REVIEW
Search an account to see score, risk band, key indicators, twelve-month repayment trajectory and bureau context.

ANOMALY REVIEW
Prioritized explanations with Confirm, Dismiss or Escalate and an analyst note.

Embedded AI controls

ASK THE PORTFOLIO
The LLM translates natural language into an allow-listed metric query. Pandas calculates every number; provenance is shown. Unsupported questions are declined.

Trigger: analyst question.
False positive: wrong query mapping.
False negative: valid question declined.
Human step: verify figures/provenance.

ANOMALY DETECTION
Isolation Forest plus peer rules. False positives are legitimate unusual behaviour; false negatives resemble normal peers. Analyst disposition is mandatory.

Model & evidence

MODEL
Selected: Logistic Regression
Customer-grouped holdout · months 1–10 only
AUC 0.679 · KS 0.285 · Gini 0.359
Precision 0.233 · Recall 0.733 · F1 0.353
Logistic retained over HGB, LightGBM and the 0.296-KS blend because it has stronger PR-AUC, recall and F1.

TOP PORTFOLIO SIGNALS
1. Lowest-income quintile default: 25.5% vs 9.5% highest-income — prioritize affordability monitoring.
2. Unemployed default: 35.0% — target early outreach.
3. Mortgage default: 24.7% — review product controls and pricing.

Assumptions and production needs

Assumptions: synthetic snapshot; monitoring/classification use case rather than origination underwriting; bureau fields treated as available at scoring; invalid collection chronology excluded from modelling; analyst decisions persist only for the demo session. Bank IT/data requirements: governed feature timestamps and outcome horizon; data contracts and lineage; authenticated role-based access; durable review/audit database; model registry and approval; scheduled scoring; encryption and secrets management; consent/retention controls; fairness review; drift, calibration, data-quality and cost monitoring; incident and rollback procedures.